

Analytical Problem Solving

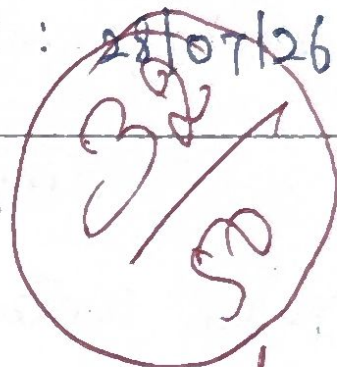
CO-2 AT-1

Name: G. Mothir

Reg no: 192421135

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vision

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1) Gray Level Transformation

= a) Gray level transformation is used to improve the visual appearance of an image values it helps in

- * Enhancing image contrast
- * Highlighting important details
- * make dark region brighter or brighter region darker.

b) Negative transformation

Formula: $S = 255 - r$

where

$r \rightarrow$ original pixel value

$S \rightarrow$ transformed pixel value

Example

original

50

100

150

200

Negative pixel

205

155

105

55

how 9

The negative image
Dark areas become bright.

② Histogram Equalization

Improve image contrast by redistributing pixel intensities over the entire intensity range.

Features:

- * Enhance low contrast images
- * Improve visibility of image details
- * Utilize the complete grayscale range

b) Cumulative Distribution function

intensity

Frequency

0

4

1

10

2

12

3

14

~~Formula~~
PDF

$$\text{Total pixel} = 4 + 10 + 14 + 12 = 40$$

3) Gaussian Smoothing

a) .

- * Reduce noise smoothly

- * Give higher weight to nearby pixel

- * preserve edges then simple averaging

b) Gaussian kernel

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\text{Pixel} = \frac{\sum (\text{kernel} \times \text{image pixel})}{16}$$

$$\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$$

$$\begin{aligned} & (1 \times 10) + (2 \times 20) + (1 \times 30) + \\ & (2 \times 40) + (4 \times 50) + (2 \times 60) + \\ & (1 \times 70) + (2 \times 80) + (1 \times 90) \end{aligned}$$

$$= 800$$

$$\text{Pixel} = \frac{800}{16}$$

$$= 50$$

Noise is reduced and the image become smoother
maintaining important image feature

④ Edge Detection Using Sobel operator

a) The edges detect the duplicate between object & the image.

- * Object detection
- * Pattern recognition
- * To verify the image

b) ~~Linear or Gradient~~ Horizontal Sobel mask

$$G_x = \begin{bmatrix} -1 & -2 & 1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$G_x = \sum (\text{mask} \times \text{image})$$

$$G_x = (-1 \times 10) + (0 \times 20) +$$

$$(1 \times 10) + (0 \times 50) + (0 \times 60) +$$

$$(0 \times 70) + (0 \times 80) + (0 \times 90)$$

$$= 80$$

$$G_x = 80$$

Interpretation: Large gradients $\rightarrow$ Strong edge

Small gradients $\rightarrow$ Smooth region

(5) Contrast Stretching

Benefits:

- by linear constant stretching

New Orange · 0-255

79:

$$Y = 100 \quad g = 50 \times 255 / 100 = 127.5 \approx 128$$

$$r = 150$$

original new

Intervention:

500 ✓	0
100	128
190	255

It use full intensity orange (0-255) producing higher contrast image.